

Portfolio Optimisation *with* Derivatives & Structured Products

BEYOND MEAN–VARIANCE

Mental Accounting Framework



From my **MSc Finance thesis** on mental-account portfolio optimisation with derivatives and structured products — and its original **R-based optimisation tool** — to a **scalable Monte-Carlo + CVaR engine**. An interactive research optimiser that builds goal-based portfolios across **four precision modes** and **four risk-based approaches**, prices derivatives and structured products, back-tests out-of-sample, and now scales to large portfolios of **many assets and many derivatives** at once.

4 risk-based methods

VaR · Expected-Shortfall · rigorous-ES · Monte-Carlo CVaR — with configurable parameters that set the risk appetite & implied risk-aversion coefficient

16 instruments

options, spreads, collars, capital-guaranteed & barrier notes

≈ 60× faster

Turbo solver — one of four grid resolutions

Multi-derivative

several derivatives optimised jointly · scales O(S·N)

10,000+ tickers

live market data — equities, ETFs & crypto

AI-powered

natural-language explanations & Q&A on every input

01 The foundations

Built on two seminal papers, both replicated in my [2012 MSc thesis](#) (USI Lugano):

- ◆ **Das, Markowitz, Scheid & Statman (2010)** — the **mental-accounts framework**: sub-portfolios as goals, each with a threshold H and shortfall probability α ; the mean-variance $\leftrightarrow$ mental-accounting **equivalence** and an implied risk aversion per goal
- ◆ **Das & Statman (2009)** — the optimisation algorithm for **derivatives & non-normal returns** in mental accounts, tested on **9 structured instruments**
- ◆ My thesis **replicated both methodologies** and extended the research to further optimisation scenarios — varying instrument parameters to confirm the equivalence and implied risk aversions, and to show the **potential benefits of adding complex instruments with non-normal return distributions**

Selected figures from my thesis — implied risk-aversion surface, copula dependence, and the bivariate-normal return density.

02 The new automated optimiser

My R thesis tool is now re-built and fully automated in **Python**: an **exact grid optimiser** that runs every simulation — on the thesis inputs or **real market data** — reproduces all the thesis results, and adds **AI-powered explanations** throughout.

- ◆ **Three risk approaches** — VaR, thesis-faithful ES, and a rigorous ES mode each with a configurable loss threshold and probability of loss
- ◆ **Four precision modes** — Fast / Standard / High + a custom **Turbo** solver (~60×) from a 21×15 grid (Fast) up to 51×99 (High) — over 20 billion state combinations at full resolution
- ◆ Enhances the optimisation by up to **+2.4 pp** of return for the same risk level

OPTIMISER OUTPUT · SAMPLE RUN

03 Extended: Derivatives & structured products

Extended the library to **sixteen instruments** with adjustable parameters, priced consistently via Black-Scholes — vanilla options, **collars**, **capital-guaranteed & barrier notes**, **spreads**, **certificates** — plus a **builder to configure your own**.

Payoff · safety collar

Priced arbitrage-consistently — and you can build and configure new instruments. The same engine drives the optimiser, the back-test and the Monte-Carlo tab.

04 Added: Out-of-sample back-test

Does the optimisation hold up out-of-sample? Build optimal weights on a construction window, then **buy-and-hold** across an evaluation window — marking the derivative **to market** each step — to **test the efficiency of the optimisation methods**.

Out-of-sample buy-and-hold performance

Same portfolio held out-of-sample, with vs without a protective put — here the put cushions the drawdowns and finishes ahead. You can back-test any instrument over any window.

05 ★ NEW — Monte-Carlo + CVaR optimiser, built for institutional-scale portfolios

The grid is exact but its state space grows as m^N — impossible past ~4 assets. The new engine instead **samples S joint scenarios** through a copula and solves the goal as a **linear program**.

- ◆ Cost is **linear in the number of assets** — and free in the number of scenarios
- ◆ **Several derivatives at once**, even on different underlyings
- ◆ Gaussian or **Student-t copula** for tail dependence (assets crash together)
- ◆ Coherent **Expected-Shortfall** floor via Rockafellar–Uryasev

max $E[r]$ s.t. $\zeta + 1/(\alpha S) \sum z_i \leq -L, \quad z_i \geq -w \cdot R_i - \zeta, \quad \sum w = 1$

The result: a **goals-based optimiser at institutional scale** — dozens of holdings and several overlays priced consistently, solved in a single, warm-started linear program.

Return / Tail-Risk Frontier · Monte-Carlo + CVaR

A representative six-asset run — AAPL, MSFT, JPM, TLT, XLE, GLD plus a protective AAPL put, 10,000 Gaussian-copula scenarios. Each dot is the maximum expected return achievable for one Expected-Shortfall floor L; ★ marks the chosen optimum (L = -20%, $\alpha = 5\%$).

OPTIMAL WEIGHTS · SCALABLE CVAR OPTIMUM

AAPL

72.8%

MSFT

22.7%

Protective put · AAPL

4.5%

$E[r]$ 24.6% · realised ES -20.0% (α 5%) · vol 25.5% · skew +0.30. JPM / TLT / XLE / GLD priced 0 at this floor — the put on AAPL lets the optimiser hold the high-return tech sleeve within the tail-risk budget.

06 Validated, not just built

- ◆ On the thesis case, Monte-Carlo optima sit **on the exact grid mean-variance frontier to ~2-3 bp**
- ◆ Scenario Expected-Shortfall matches the **closed-form Normal value** (e.g. -28.4% vs -28.6%)
- ◆ The frontier is **monotone**, the ES floor **binds exactly**, and infeasible floors are flagged
- ◆ Approximate-but-scalable engine that **complements** the exact reference — never replaces it

07 Under the hood

A clean pipeline: **scenarios** → **CVaR linear program** → **warm-started frontier**, wrapped in an interactive app with live charts, weight bars and back-testing.

Python

NumPy

SciPy · HiGHS LP

Streamlit

Plotly

Altair

Copulas

Black-Scholes

Rockafellar–Uryasev CVaR

SJ

Sami Jeddou

Senior Financial Services Transformation — Risk & Capital Markets

Interactive Python · Streamlit app — [user guide available](#)

Happy to discuss with asset managers & quant teams — feel free to connect

Research & educational project — not investment advice. Builds on Das, Markowitz, Scheid & Statman (2010) and my 2012 MSc Finance thesis (USI Lugano).